

Unit ET460

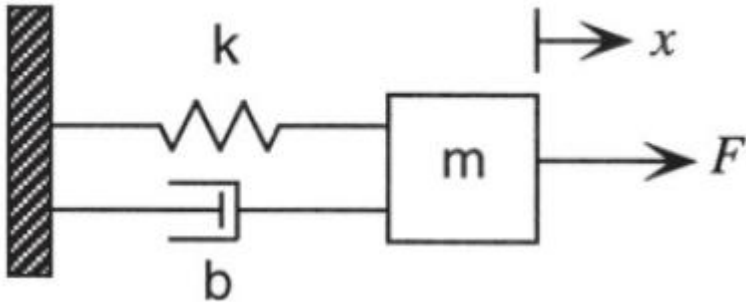
Unit 1 - Tutorial

Differential Equation

Mechanical

$$m \cdot \frac{d^2}{dt^2} x(t) + b \cdot \frac{d}{dt} x(t) + k \cdot x(t) = 0$$

$$m \cdot \ddot{x} + b \cdot \dot{x} + k \cdot x = 0$$

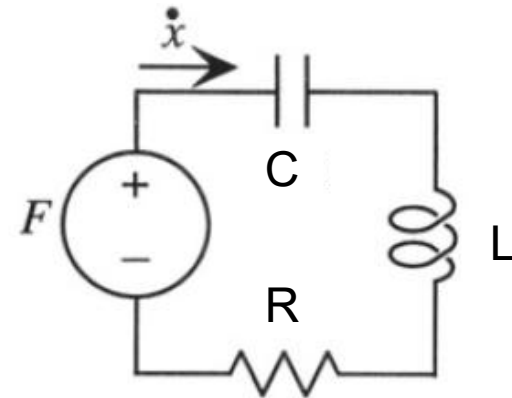


Electrical

$$L \cdot \frac{d^2}{dt^2} q(t) + R \cdot \frac{d}{dt} q(t) + \frac{1}{C} \cdot q(t) = 0$$

$$L \cdot \ddot{q} + R \cdot \dot{q} + \frac{1}{C} \cdot q = 0$$

$$L \cdot \dot{I} + R \cdot I + \frac{1}{C} \cdot \int I dt = 0$$



Equivalent circuit

Damping Ratio/Factor

$$\zeta = \frac{b}{c_c} = \frac{\text{actual damping}}{\text{critical damping}}$$

general

$$m \cdot \ddot{x} + b \cdot \dot{x} + k \cdot x = 0$$

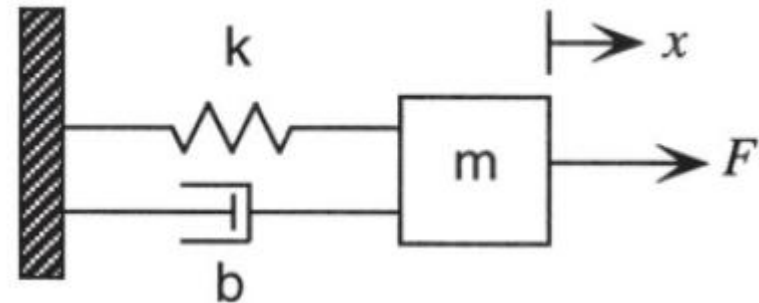
universal description
of system behavior

system energy related

$$c_c = 2\sqrt{k \cdot m}$$

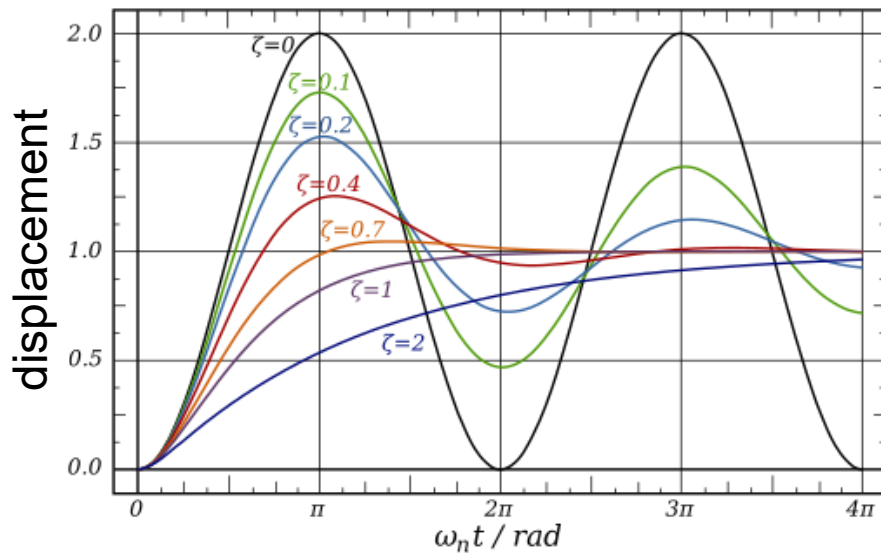
attenuation or
decay rate

$$\zeta = \frac{b}{2\sqrt{k \cdot m}} = \frac{\alpha}{\omega_0} = \frac{b/2m}{\sqrt{k/m}}$$

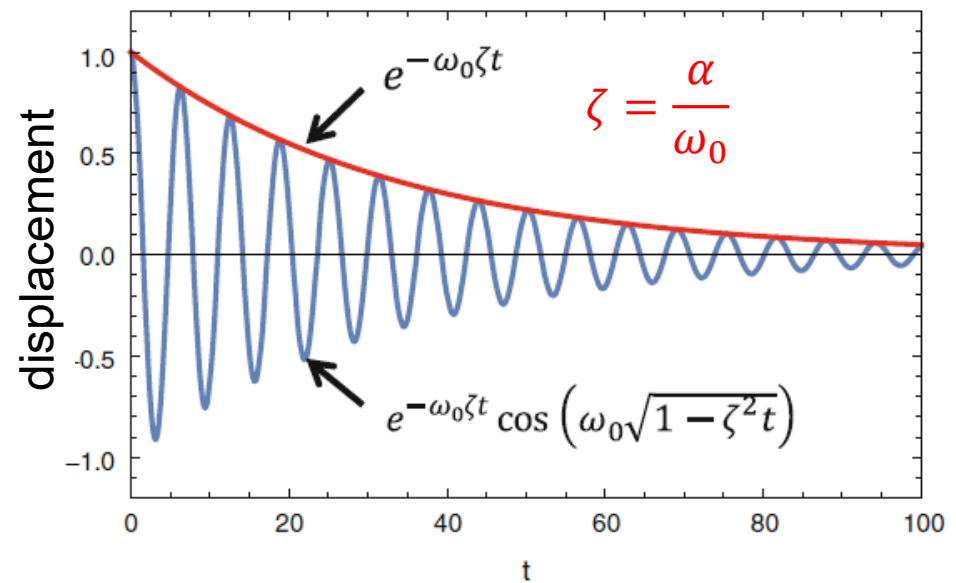


Damping and Attenuation

step response

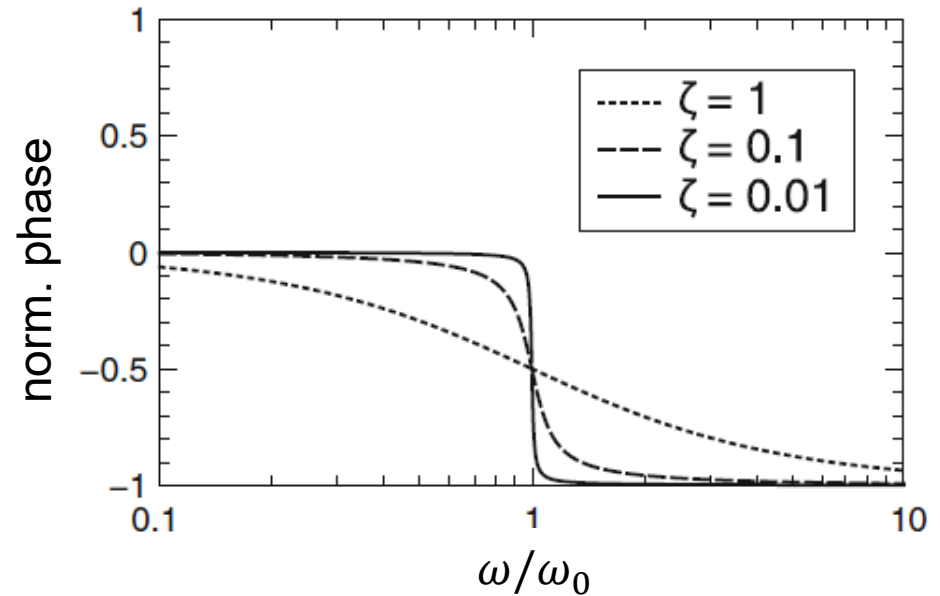
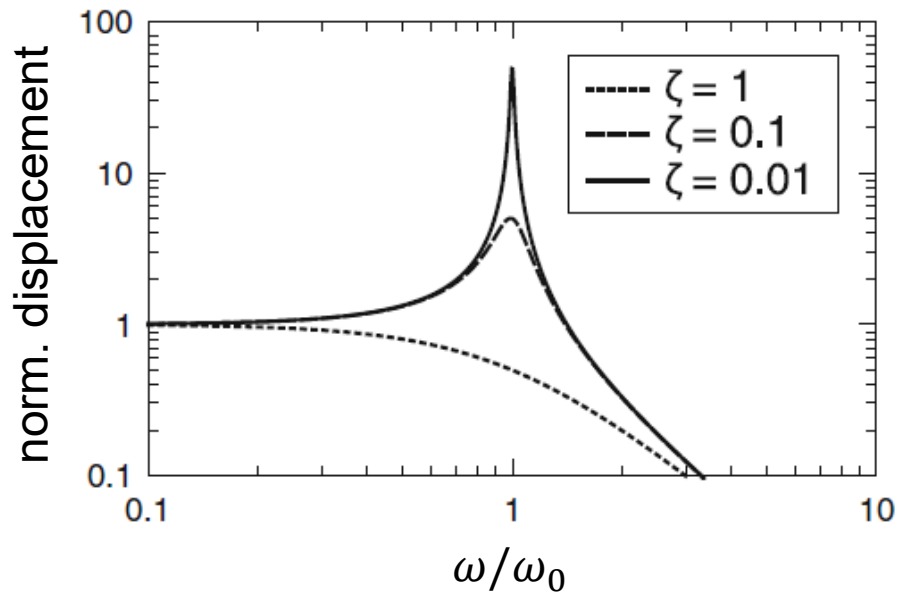


free oscillation



Damping and Attenuation

frequency response



Quality Factor

$$Q = 2\pi \cdot \frac{\text{energy stored}}{\text{energy loss per cycle}}$$

$$\zeta = \frac{b}{c_c} = \frac{\text{actual damping}}{\text{critical damping}}$$

$$Q = \frac{1}{2\zeta} = \frac{\omega_0}{2\alpha}$$

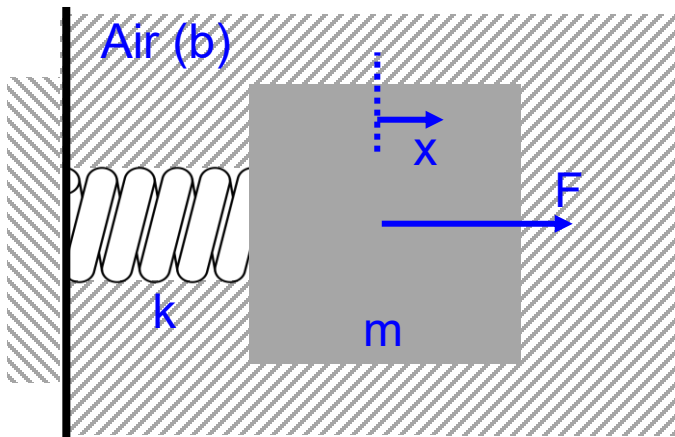
$$Q = \frac{1}{2 \cdot \frac{b}{2\sqrt{k \cdot m}}} = \frac{\sqrt{k \cdot m}}{b}$$

Damping Cheat Sheet

parameter	meaning	trans. mechanical	electrical RLC
natural frequency eigenfrequency	rate at which a system tends to oscillate in absence of disturbance and without damping	$\omega_0 = \sqrt{\frac{k}{m}}$	$\omega_0 = \frac{1}{\sqrt{LC}}$
resonance frequency	rate at which a system tends to oscillate in absence of disturbance but with damping	$\omega_r = \omega_0 \cdot \sqrt{1 - \zeta^2}$	$\omega_r = \omega_0 \cdot \sqrt{1 - \zeta^2}$
decay rate attenuation Neper frequency	rate of decay of the amplitude of a free oscillation	$\alpha = \frac{b}{2m}$	$\alpha = \frac{R}{2L}$
damping ratio	“how damped” a system is, generally describes the behavior	$\zeta = \frac{\alpha}{\omega_0} = \frac{b}{2\sqrt{km}}$	$\zeta = \frac{R}{2} \sqrt{\frac{C}{L}}$
quality factor Q factor	how underdamped a system is/how well it stores energy	$Q = 2\pi \cdot \frac{\text{energy stored}}{\text{energy lost per cycle}}$ $Q = \frac{1}{2\zeta} = \frac{\sqrt{km}}{b}$	$Q = \frac{1}{R} \sqrt{\frac{L}{C}}$

Example

- Mass of attached to a spring of with air damping



$$\begin{aligned}m &= 0.1 \text{ kg} \\k &= 100 \text{ N/m} \\\zeta &= 0.01\end{aligned}$$

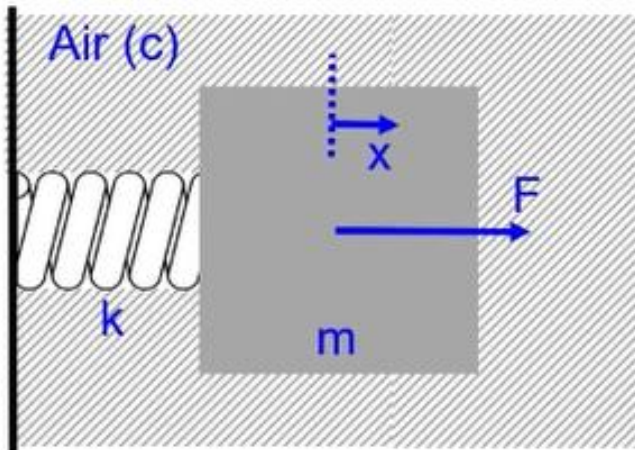
Analysis:

- What is the resonance frequency?

Simulation:

- build electrical equivalent model
- simulate and plot the frequency response of the **flow variable**
 - 0.1 to 100 Hz
 - what is the electrical/mechanical meaning of the flow variable?
- plot the frequency response of the **displacement x** for the case that the system is
 - underdamped
 - critically damped
 - overdamped

Mass-Spring-Damper



2nd Newton's law:

$$F(t) - kx(t) - c \frac{dx(t)}{dt} = m \frac{d^2x(t)}{dt^2}$$

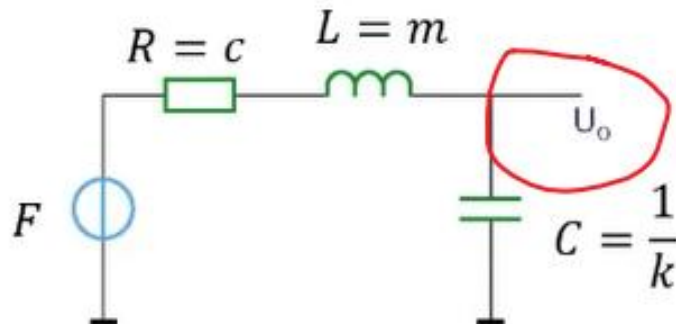
Rearranging, and taking the Laplace transform:

$$\frac{X(s)}{F(s)} = \frac{1}{ms^2 + cs + k}$$

displacement

$$\frac{V(s)}{F(s)} = \frac{1}{ms + c + \frac{k}{s}}$$

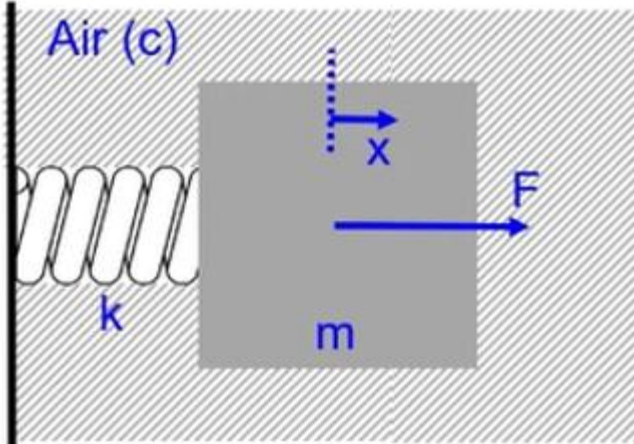
velocity



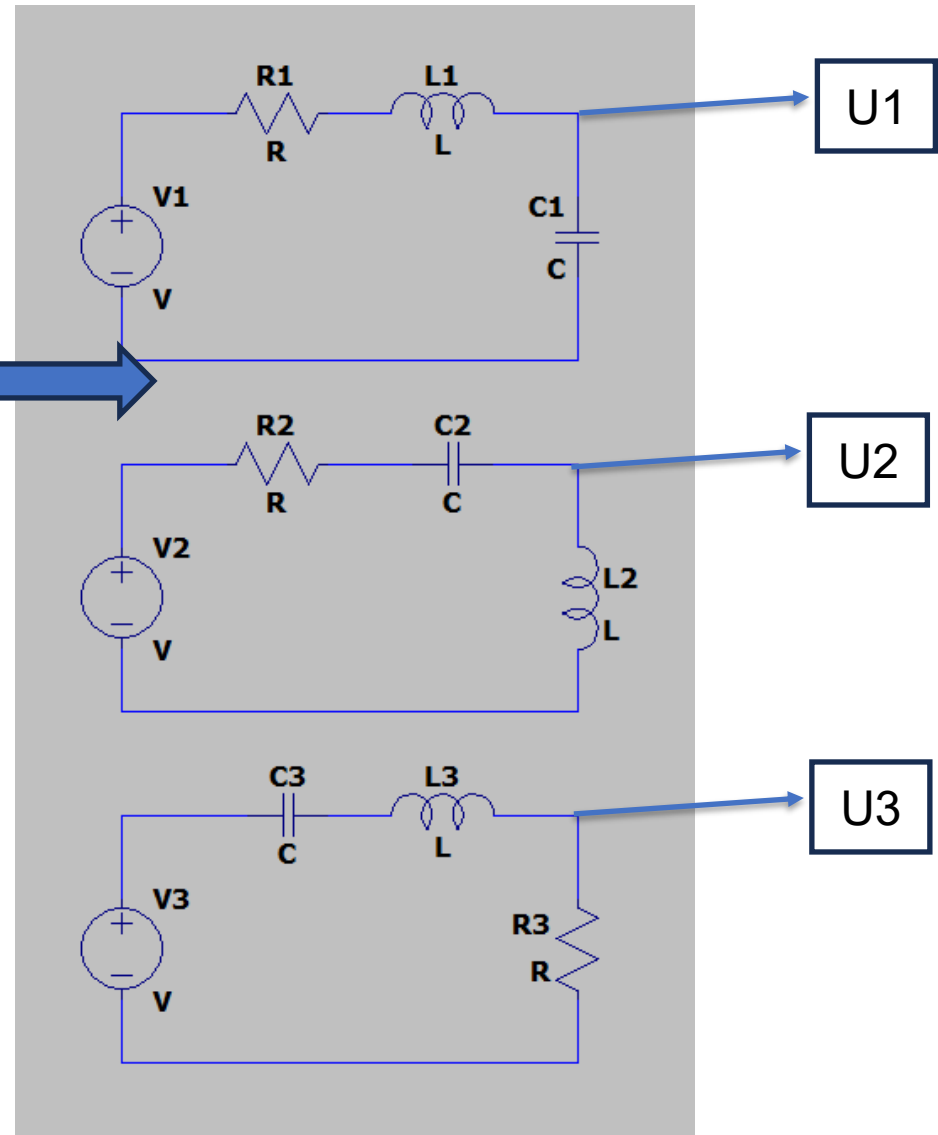
$$\frac{I(s)}{U(s)} = \frac{1}{Ls + R + \frac{1}{Cs}}$$

RLC circuit (2nd order)

Circuit Model

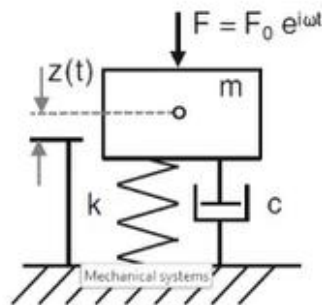


Which can be used to extract the displacement response $x(t)$?

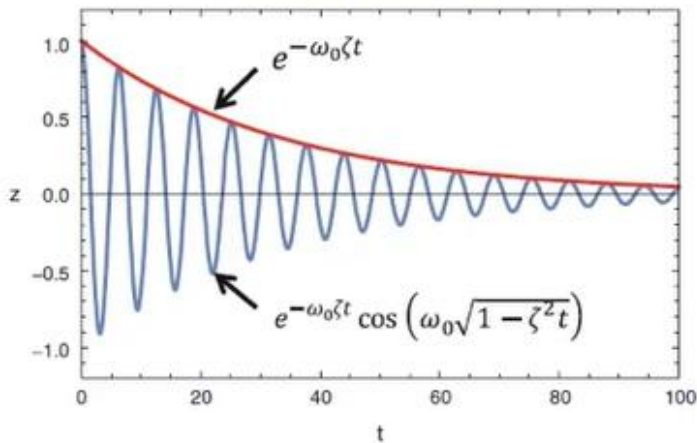


2nd order systems

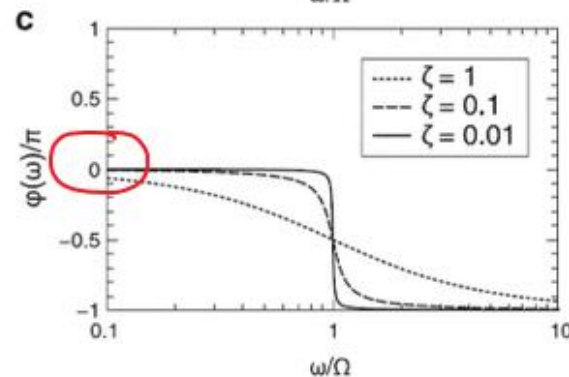
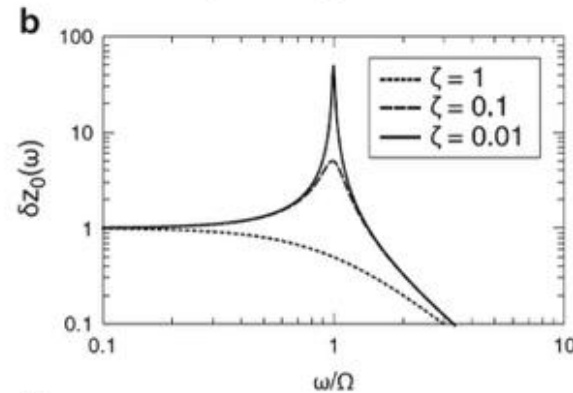
Physically, why does the displacement phase shift from 0° to -180° as the driving frequency increases?



Time domain



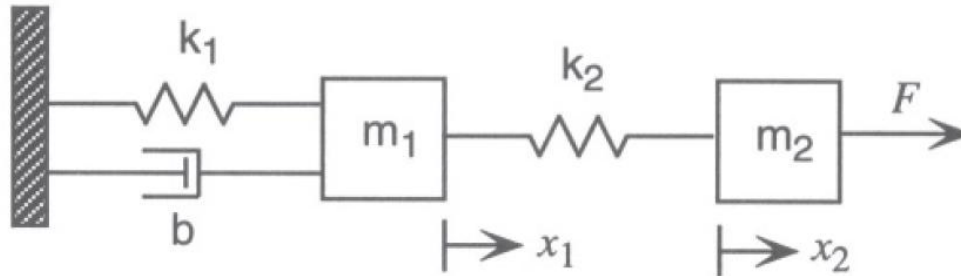
Frequency domain



$$\frac{X(s)}{F(s)} = \frac{1}{ms^2 + cs + k}$$

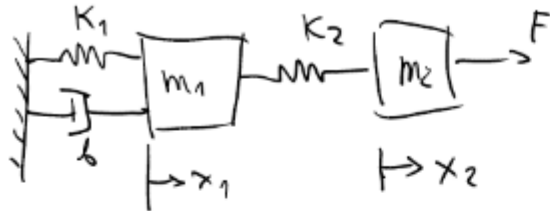
Example

- Find the equivalent electrical circuit of this mechanical system



Example

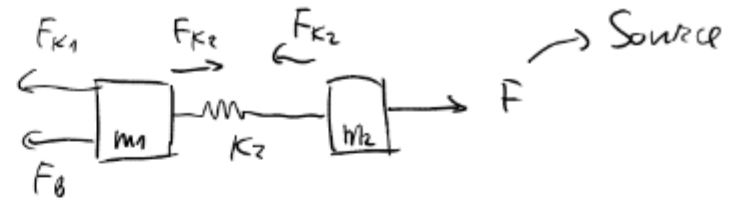
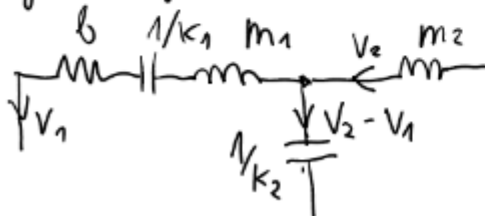
Exercise 1:



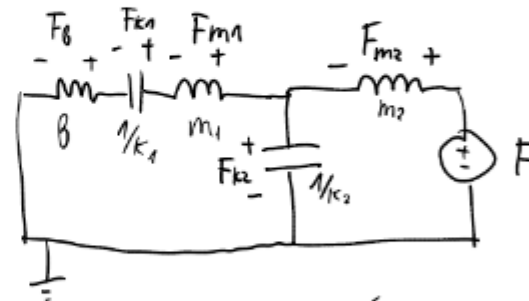
- K_1 , b and $m \rightarrow$ Same velocity $\rightarrow$ In series



- Flow of K_2 is the difference between flows of m_1 and m_2 .



- Force on m_2 is equal to external force (source), minus force on K_2

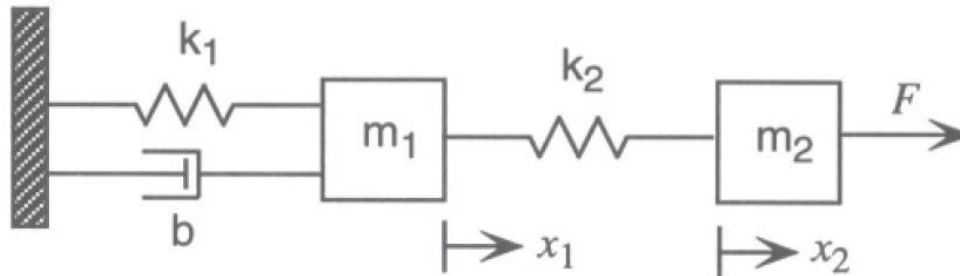


$$F_{m2} = F - F_{K2} \quad \checkmark$$

$$F_{m1} = F_{K2} - F_{K1} - F_b \quad \checkmark$$

Example

- Simulate the step response of x_1 in LTSpice



$$k_1 = 1 \text{ N/m}$$

$$m_1 = 2 \text{ kg}$$

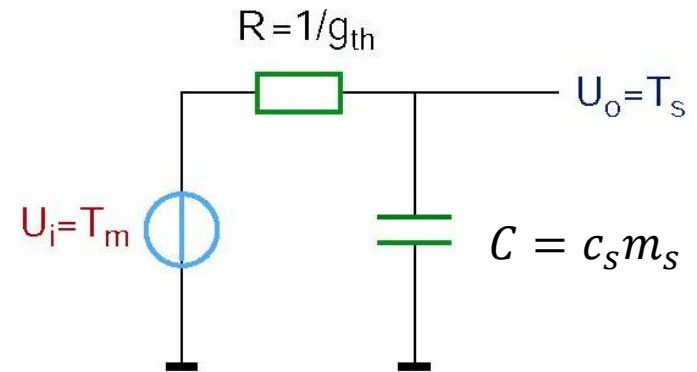
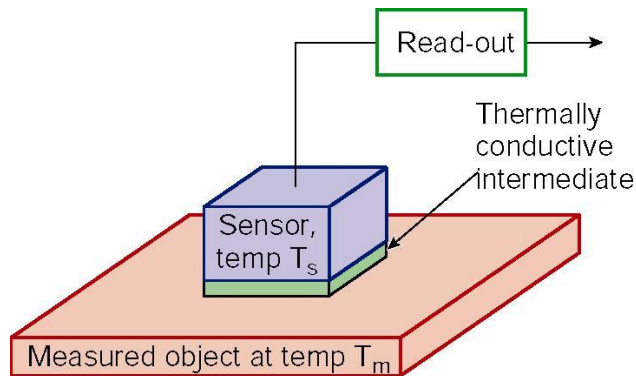
$$b = 5 \text{ Ns/m}$$

$$k_2 = 3 \text{ N/m}$$

$$m_2 = 1 \text{ kg}$$

Example: Temperature sensor

- Can we represent it as a circuit?



$$\frac{T_s(s)}{T_m(s)} = \frac{1}{1 + \frac{c_s m_s}{g_{th}} s}$$

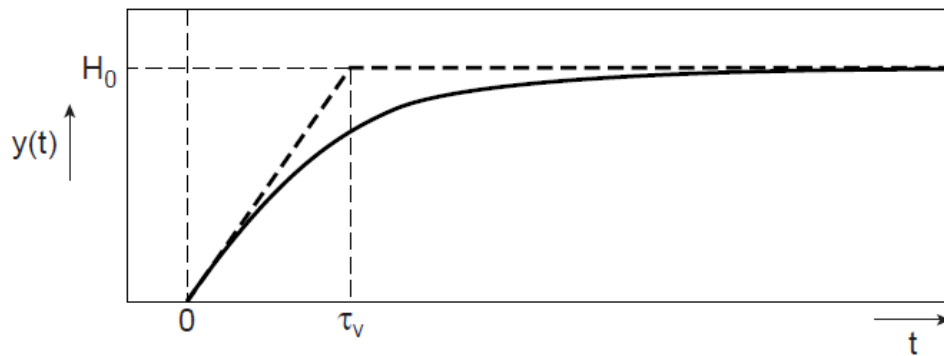
$$\frac{U_o(s)}{U_i(s)} = \frac{1}{1 + RCs}$$

Input	T_m	U_i
Output	T_s	U_o
Time constant	$c_s m_s / g_{th}$	RC

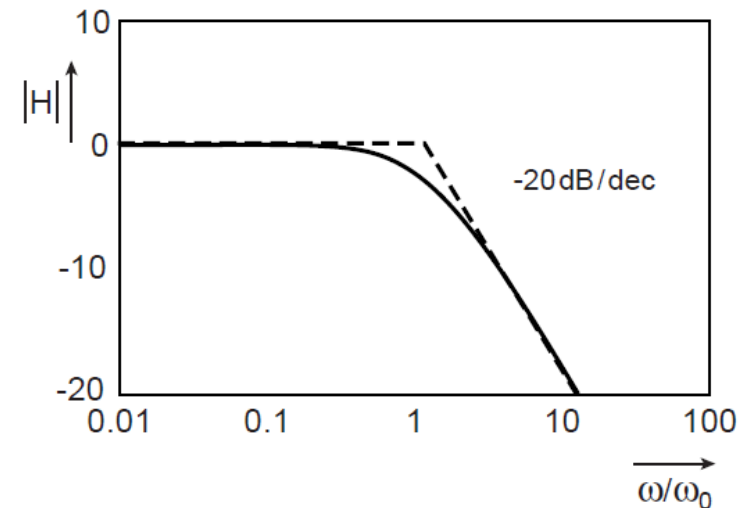
Example: Temperature sensor

- You can apply everything you know about 1st order circuits/systems

Time domain

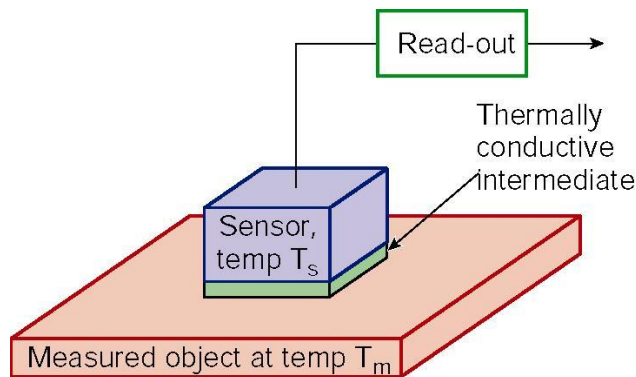


Frequency domain



Example

- Which of these sensors takes longer to reach the temperature of the substrate?
 - A. Sensor 1
 - B. Sensor 2
 - C. It takes the same time for both.



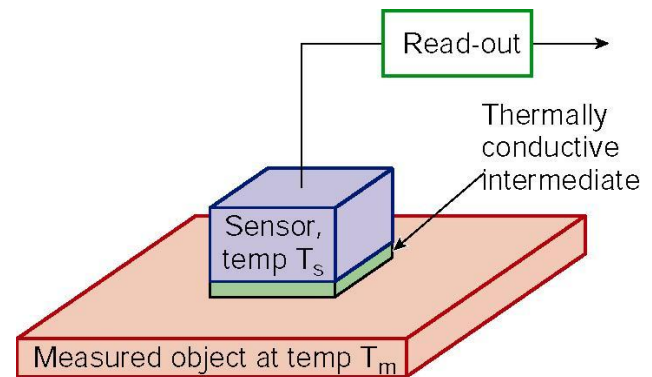
Sensor 1:

Sensor volume: $10 \times 10 \times 10 \text{ mm}^3$

Density of sensor material: 1500 kg/m^3

Specific heat of sensor material: 700 J/(kg K)

Conductivity of intermediate material: 500 W/K



Sensor 2:

Sensor volume: $10 \times 10 \times 10 \text{ mm}^3$

Density of sensor material: 3000 kg/m^3

Specific heat of sensor material: 700 J/(kg K)

Conductivity of intermediate material: 250 W/K

Example

- Which of these sensors takes longer to reach the temperature of the substrate?
 - A. Sensor 1
 - B. Sensor 2**
 - C. It takes the same time for both.

Time constant of sensor temperature:

$$\tau_c = \frac{c_s m_s}{g_{th}} = RC$$

$$\begin{array}{l} m_{s2} = 2 m_{s1} \\ g_{th2} = 0.5 g_{th1} \end{array} \quad \Rightarrow \quad \tau_{c2} = 4 \tau_{c1}$$

LTSpice Simulation

Sensor 1:

Sensor volume: $10 \times 10 \times 10 \text{ mm}^3$

Density of sensor material: 1500 kg/m^3

Specific heat of sensor material: 700 J/(kg K)

Conductivity of intermediate material: 500 W/K

$$c_{s1} m_{s1} = 1.05 \text{ J/K}$$

$$1/g_{th1} = 0.002 \text{ K/W}$$

Sensor 2:

Sensor volume: $10 \times 10 \times 10 \text{ mm}^3$

Density of sensor material: 3000 kg/m^3

Specific heat of sensor material: 700 J/(kg K)

Conductivity of intermediate material: 250 W/K

$$c_{s2} m_{s2} = 2.1 \text{ J/K}$$

$$1/g_{th1} = 0.004 \text{ K/W}$$

